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Innovation Guide for AI Agents

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AI agents promise to make AI more autonomous and goal oriented. CIOs should understand the benefits, risks, use cases and the evolving technology ecosystem for AI agents to create a robust strategy for agentic AI.

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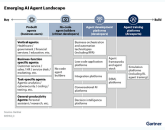

Overview

Key Findings

- There is a high degree of enterprise AI ambition around **AI agents**, which has given rise to a large number of proof-of-concepts and a strategic plan to increase their AI-related budgets over the next year.
- However, according to the Gartner CIO survey, only 16% of organizations have deployed **agentic AI**. Meanwhile, greater than 80% of IT leaders believe their enterprise data is not ready for agentic AI, while over 70% worry about **AI governance** with the proliferation of AI agents.
- The AI agent technology ecosystem is emerging rapidly with domain specific agents, low code and no code agent platforms, agent development frameworks and a number of platform tools for AI agent

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training. SaaS vendors are aggressively embedding AI agents into enterprise software as they seek to maintain their incumbency.

- While there has been progress across several fronts, such as reasoning models and open integration protocols to enable capable agents, technologies such as **world models** and agent interfaces to the industrial world are still nascent and immature.

Recommendations

- Choose vendors that adopt open protocols like MCP, A2A to avoid deep ecosystem lock-in and ensure interoperability across agent providers.

- Avoid Ad hoc AI agent deployments that are technology driven and instead target pain points where AI agents offer measurable ROI in areas such as software engineering, **DevOps**, customer

service, and sales operations.

- Do not buy solutions based on the vendor's vision — buy based on their ability to deliver production ready solutions today and ability to demonstrate clear value through a proof-of-concept.
- Treat agents as discrete entities with clearly defined permissions, access boundaries, and oversight. Deploy tools to guard against prompt injection, goal misalignment, and rogue actions.

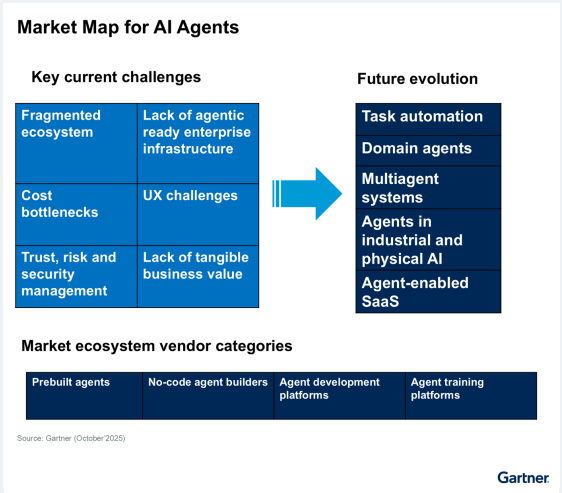
Market Definition

AI agents are autonomous or semiautonomous software entities that use AI techniques to perceive, make decisions, take actions and achieve goals in their digital or physical environments. Vendors are striving to evolve them into **multiagent systems (MAS)**, which are collections of AI agents that interact with each other to achieve

individual or shared goals. These agents can be delivered in a single environment or can be independently developed and deployed on distributed environments.

Market Map

Figure 1: Market Map for AI Agents



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Market Dynamics

- **Surge in enterprise interest and pilots:** There has been an explosion of interest in AI agents for automation in areas such as sourcing and procurement, customer support, and DevOps. However,

most deployments are in pilot or experimentation phase. 17% of the respondents to the 2026 Gartner CIO and Technology Executive Survey indicated that their enterprise had already deployed AI agents, but 42% said their enterprise would deploy within the next year (i.e., by 2026).

- **Fragmented but rapid ecosystem expansion:** Hundreds of open-source and commercial agentic frameworks have emerged, creating overload of choice and interoperability issues.

- **Infrastructure for AI agents isn't in place:** Agentic systems require persistent memory, dynamic planning, mature retrieval-augmented generation (RAG) pipelines, and tool orchestration, all of which stretch current MLOps and IT stack capabilities. Integration of AI agents with enterprise

systems, workflows and tools remains immature. Moreover, according to the Gartner Agentic AI Survey, greater than 80% of IT leaders believe their enterprise data is not ready for agentic AI while greater than 70% worry about AI governance with the proliferation of AI agents.

- **Cost, latency, and UX bottlenecks:** Agentic systems can be expensive and slow, especially with long prompt chains, tool usage overhead and multiagent orchestration. Moreover, current SaaS vendor pricing models are often not conducive to agentic AI as large vendors try to ward off the threat to their existing seat based pricing models.

- **Governance, control, and safety are enterprise blockers:** CIOs want AI agents, but agents must be:

  - Trustworthy, traceable,

auditable and interpretable

- Integrated with human-in-the-loop and capable of seamless hand-off
- Aligned with compliance and data privacy requirements

Market Evolution

- **Shift from chatbots to task-automating agents:** Enterprises are keen to move beyond generative AI (GenAI) chat interfaces toward agents that perform tasks end-to-end. This shift is driven by ROI pressure and demand for more tangible business outcomes.
- **Emergence of domain-specific agents: Agentic AI is seeing verticalization in several areas such as:**
  - IT and DevOps agents (e.g., incident management, root cause analysis)
  - Sales agents (e.g., cold

calling, proposal creation)

- Finance agents (e.g., expense audit, invoice matching)
- Customer service agents (e.g., intent-based resolution)
- Software engineering agents (e.g., code completion, code analysis)
- Deep research agents (e.g., analysis, market intelligence)

- **From single agents to multi-agent systems:** In the future, agent groups with specialized roles will handle collaborative business processes. Companies like Anthropic, Google, Cisco and others are creating agentic middleware, model-agnostic control planes that sit between enterprise systems and multiple agents or models. Emergence of open-source integration projects such as MCP, A2A and ACP promise to bring open standards to agentic AI.

- **From digital to industrial AI:** While AI agents predominantly operate in the digital world, the convergence of AI agents with robotics and industrial systems will enable use cases that harness the power of natural language, computer vision with actions.

- **Agent enabled software-as-a-service (SaaS):** SaaS apps will evolve into more autonomous agent services, where users specify intent (e.g., "Plan our next campaign"), and the system uses agents to execute it end-to-end. This will require a shift from User UX — such as, clicks, natural language input — to APIs, function calls and memory tools.

Business Benefits and Use Cases

The key business benefits of AI agents are:

- **Productivity gains:** Automates

multistep, cross-functional tasks and reduces manual work and cognitive load, freeing employees for high-value activities.

- **Always-on digital resource:** Operates 24/7 across time zones, ensuring continuous task execution, monitoring, and exception handling.
- **Lower total cost of operations:** Reduces need for manual workflows, BPO, or rigid RPA automations.
- **Enhanced customer and employee experience:** Enables personalized, responsive interactions and seamless workflows across apps, channels, and teams.

Use Cases:

- **Workflow automation and orchestration:** AI agents can sit atop ERP, CRM and BI platforms to trigger and manage business processes such as procure-to-

pay, cold calling, with limited human orchestration.

- **Customer service and support automation:** Agentic AI can triage incoming requests, pull context from CRM records, and resolve support issues.
- **IT service management and DevOps:** Agentic AI enables self-healing IT operations by detecting incidents, performing root cause analysis, generating remediation runbooks, and executing automated fixes.
- **Finance andaccounting:** Agentic AI can handle expense reporting, perform real-time cash-flow forecasting, run scenario analyses, and propose budget adjustments.
- **HR and talent management:** Agentic AI can schedule interviews, autonomously coordinate calendars, trigger training modules, and flag retention risks based on employee feedback and

performance data.

- **Salesand marketing operations:** Agentic AI can identify high-value leads, execute personalized outreach sequences, and continuously optimize campaigns by analyzing engagement metrics.
- **Product and software development:** AI coding agents can generate boilerplate and production-ready code, review pull requests, accelerate prototyping and reduce time-to-market. They can also run test suites, triage bugs, and improve product quality.
- **Compliance, risk and security:** AI agents continuously monitor logs and network traffic, detect anomalies or policy violations, and trigger investigation workflows or automated containment.
- **Deep research:** Deep research agents help decision makers rapidly synthesize complex data

and insights across sources to make faster, more informed strategic decisions.

Managing Risks

AI agents bring about new risks beyond those associated with traditional AI models and applications given their increased autonomy. In standard AI systems, the threat landscape typically encompasses inputs, model processing, outputs, software vulnerabilities, and the insecure environments hosting them. However, with AI agents, this threat landscape broadens to include the sequences of events and interactions they initiate and participate in — activities that are inherently invisible to and uncontrollable by human operators. Some of the key risks include:

- **End-to-end data exposure risks:** Data can be unintentionally exposed or exfiltrated at any stage of an AI agent's life cycle,

making holistic security monitoring essential.

- **Agent sprawl:** Given the proliferation of low-code and no-code agent platforms, the risks of unmanaged, ungoverned agent sprawl is a real risk for enterprises. This is also compounded by the lack of multiagent control planes for service discovery and life cycle management of agents.
- **Agent misuse and hijacking:** Autonomous agents can be hijacked or manipulated by external threats or internal errors, leading to unauthorized or harmful actions. This includes attacks where agents are reprogrammed by malicious actors or poisoning of retrieval-augmented generation (RAG) pipelines that influence agent behavior.
- **Logic flaws and unintended behaviors in agent code:** Agents may generate or execute

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flawed logic that leads to data breaches, compliance violations, or system vulnerabilities — even without explicit malicious intent.

- **Third-party supply chain vulnerabilities:** Agents often leverage open-source libraries or third-party components, introducing risk of malware or compromised code infiltrating both traditional and AI-driven systems.
- **Unreliable outputs due to hallucinations:** Autonomous AI agents may generate factually incorrect or fabricated information with high confidence, leading to poor decision-making, compliance violations, or reputational harm if left unchecked in business-critical workflows.
- **Embedded and amplified bias:** AI agents can inherit or amplify biases present in training data or prompts, resulting in unfair,

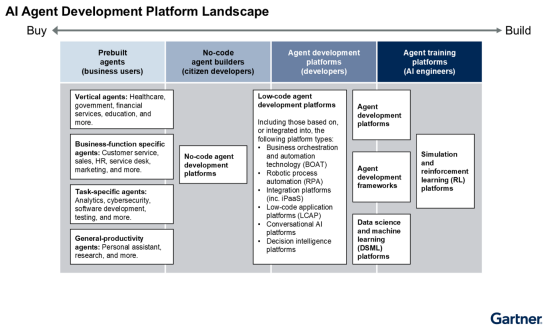
discriminatory, or unethical behavior, particularly problematic in domains like HR, finance, customer service, or legal advisory functions.

- **Unintended escalation or decision loops:** Hallucinations and flawed reasoning can cause agentic AI to pursue erroneous goals, escalate issues unnecessarily, or initiate inappropriate actions, creating operational disruptions or policy violations.

Representative Vendors

AI Agent Technology Landscape

Figure 2. AI Agent Technology Landscape



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The AI agent vendor landscape is broad and rapidly evolving, with a lot of vendors claiming AI agent capabilities. Like most early stage markets, it is characterized by tall claims, hyper marketing and rapid evolution. Here are some of the key AI agent vendors across the approaches described above. This isn't meant to be a comprehensive compendium of vendors but a bird's eye view of key emerging vendors in a nascent market.

Table 1: Vertical Agents

Industry	*Company name*
Healthcare	Hippocratic AI, Thoughtful AI, Assort Health, Synthpop
Legal	Harvey AI, Spellbook
Transportation and logistics	Fleetworks, Didero, HappyRobot

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Table 2: Business Function Agents

Enlarge Table

Business function	*Company name*
HR	Apriora, Moonhub, Borderless AI
Marketing	Scrunch AI, Ability AI, Firsthand AI

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Table 3: Task Agents

Business function	*Company name*	*Us foc*
Data analysis	Athena Intelligence, Paradigm, Seek AI, Snowflake, Wisdom AI	Dat ana anc nat lan to

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Table 4: Knowledge Management, Search and General Productivity Agents

Business function	*Company name*	
Search, knowledge management and productivity	Anthropic, OpenAI, Perplexity, Glean, Google, Microsoft, Notion, Salesforce (Slack)	

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No Code Agent Development Platforms:

This includes a broad set of no-code AI agent solutions that provide a development and deployment platform for creating, orchestrating and deploying agents across a variety of use cases and scenarios.

Table 5: No-Code Agent development platforms

Description	*Sample vendors*
No-code agent builders are products that offer an integrated design and runtime environment to build, publish and manage AI-powered agents without exposing any code to the builder.	Asana, aiXplain, Dify AI, Ema AI, Ninjatech AI

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Low-Code Agent Development Platforms

This includes a broad category of low-code agent development platforms.

Table 6: Agent Platforms by Software Categories

Business orchestration and automation technology (BOAT), vendors, vendors and platforms	***Automation Anywhere, UiPath***
Low-code application vendors	Appian, OutSystems
Conversational AI platforms	Kore AI, Leena AI
Integration platform	Boomi, Workato, Zapier
Decision	Aera

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Table 7: Agent Development Platforms

Enlarge Table

Vendor	*Description*
Microsoft Copilot Studio	A low-code platform to build AI copilots (agents) that interact through Teams, websites, or internal apps and integrated with Microsoft tools.
Salesforce Agentforce	Agentforce enables devs to create

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Table 8: Data Science and ML Platforms (DSML Platforms)

Enlarge Table

Vendor	*Description*
Databricks	Databricks provides frameworks and tooling — such as, Mosaic AI Agent Framework, vector search, DBRX models — to architect, deploy, and orchestrate agentic AI workflows across its unified platform.

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AI Agent Development Frameworks

This is a new category and emerging frameworks targeting the builders, primarily developers.

Table 9: AI Agent Development Frameworks

Enlarge Table

AI agent development framework	*Description*
AWS Strands Agents	Strands Agents is an open-source SDK from AWS that enables developers to build AI agents using a model-first approach instead of manually designing rigid workflows.

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In addition, there are high-level technical computing platforms

with tools for training AI agents, especially in control systems and engineering simulations through reinforcement learning training. Examples of such platforms include Matlab, OpenAI Gym, Ray RLib and more. In addition, there is a growing set of tools aiding enterprises in agentic operations:

Table 10: Agentic Operations Components

Enlarge Table

Agent components	*What are they?*
Fine tuning tools	Customizes and fine-tunes models for specific tasks
Prompt engineering and workflow orchestration	Manages complex agent behaviors, tools, memory
Memory	Enables contextual memory,

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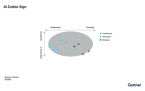

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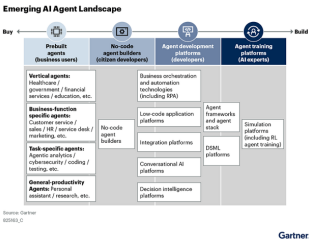

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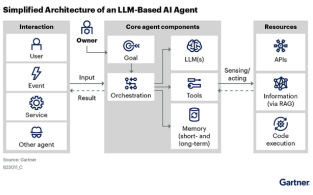

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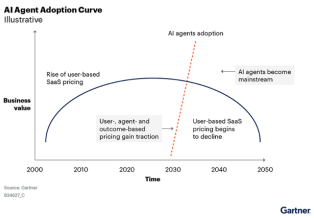

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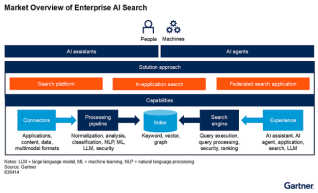

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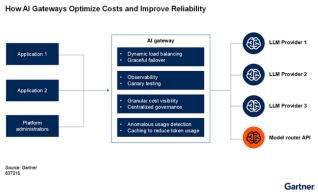

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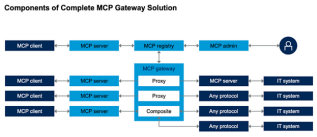

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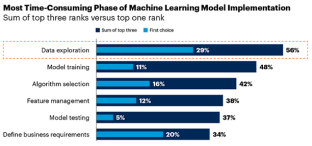

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